

R-Value: Pricing NBA Production Against True Team Expenditure

Introduction. Public player-value metrics assess production but ignore what it costs. Nominal salary badly understates that cost: under the luxury tax, a dollar paid by a repeater taxpayer can cost its franchise several times a dollar paid by a team below the line. We ask which players and teams achieve the highest production per dollar of true expenditure, and whether cost-efficiency is linked to winning championships. The question is timely: the 2023 CBA’s apron system was designed specifically to limit high-spending rosters.

Methods. R-Value pairs a performance core with a CBA-correct cost model. The core combines five z-scored pillars z_k —box production (rate and volume), usage-scaled shooting efficiency, minute-shrunk on-court offensive and defensive impact, and PIE—into a composite index, maps it through the normal CDF Φ , and adjusts for availability over m minutes:

$$\text{CPI} = \sum_{k=1}^5 w_k z_k, \quad \text{PS} = 100 \Phi(\text{CPI}), \quad \text{WPS} = \text{PS}(0.55 + 0.45A), \quad A = \sqrt{\frac{\min(m, 2200)}{2200}}.$$

The weights w_k are set by a two-stage simplex grid search maximising All-NBA recall@15 and team-win rank correlation. The cost model computes each season’s luxury tax $T(P)$ on payroll P from the bracket schedule then in effect (2017 CBA through 2022–23, the 2023 CBA thereafter), applies repeater rates, and grosses up every salary s_i by the team’s total-expenditure multiplier:

$$\text{TE} = P + T(P), \quad \text{CI}_i = \frac{s_i (\text{TE}/P)}{\text{TE}}, \quad R_i = 100 \cdot \text{norm} \left[\text{WPS}_i \left(\frac{\overline{\text{CI}}}{\text{CI}_i} \right)^{0.35} \right],$$

so that the league average is 100. Team R-Value divides minute-weighted production by total expenditure. The data covers six seasons—2018–19 and 2021–22 through 2025–26—excluding the COVID-disrupted 2019–20 and 2020–21, whose irregular schedules and pro-rated tax settlements compromise the availability and cost models respectively.

Results. Under leave-one-season-out cross-validation—every parameter refitted excluding the season being scored—the core recovers 78.9% of All-NBA selections among its top 15, against 71.1% for PIE×minutes, 34.4% for raw plus-minus and 23.3% for estimated net rating, and correlates with team wins at $\rho = 0.914$ (Table 1). Weights are stable: four of six folds choose the same vector, the offensive-impact weight identical in all six.

Championships are not usually acquired efficiently (Figure 1). For the six champions, production-per-dollar ranked 16th, 29th, 14th, 13th, 3rd and 14th of 30. The 2021–22 Warriors won at an R-Value of 49, on a \$179M payroll carrying a \$191M repeater tax. A fitted title score weights eight-man rotation quality roughly nine times the top-two star premium, and ranks actual champions 2.83rd of 30 out of sample. Across the 2025–26 season’s 582 players, R-Value is right-skewed at +1.04 (with a median of 94.9 against a mean fixed at 100): most contracts return below-average value per dollar, while a minority of rookie-scale deals return multiples of it.

Conclusion. The league’s most cost-efficient rosters are rarely the ones that win titles; teams convert brief value windows into contention by spending through them. R-Value expresses that trade-off in CBA-accurate dollars, letting front offices evaluate roster decisions against actual expenditure rather than nominal salary, and ranks players, rates teams and forecasts seasons from one model.

Code and data: [\[GitHub repository\]](#)

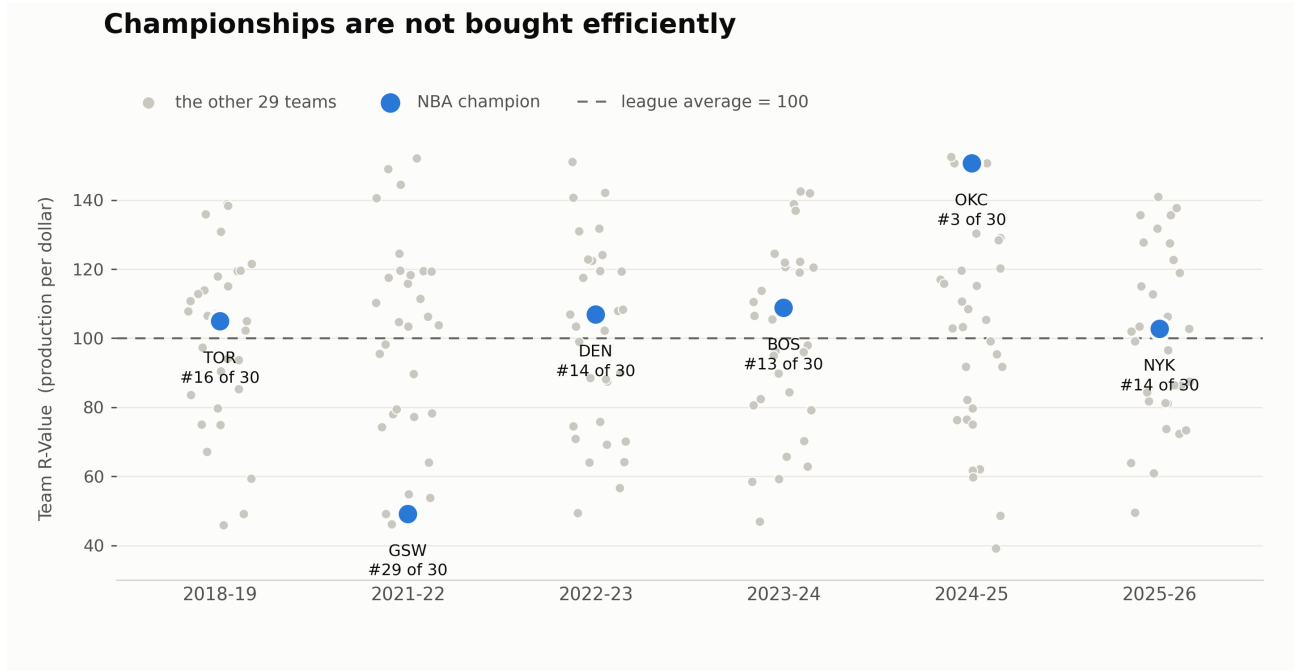


Figure 1: Team R-Value for all 30 teams across six seasons, champion highlighted. Five of the six champions sit at or below the league average of 100.

Table 1: Leave-one-season-out cross-validation. Every parameter (five pillar weights, two title premiums) is refit on the other five seasons, then scored on the held-out season.

Held-out season	R-Value recall@15	ρ with wins	Champion title rank
2018-19	0.800	0.935	1
2021-22	0.800	0.882	3
2022-23	0.733	0.880	6
2023-24	0.800	0.926	1
2024-25	0.800	0.932	3
2025-26	0.800	0.927	3
Mean (out-of-sample)	0.789	0.914	2.83 of 30

Benchmarks, which carry no fitted parameters: PIE×minutes 0.711, raw plus-minus 0.344, estimated net rating 0.233.